

MODULE ONE

CONCEPT OF ARTIFICIAL INTELLIGENCE AND TOOLS

1.1 ARTIFICIAL INTELLIGENCE (AI):-

Is the advance field of computer science dedicated to automating behavior commonly associated with human intelligence. AI is the simulation of human intelligence process by Machine, especially computer systems. Specific application of AI include experts systems, natural language processing and speech recognition and machine learning.

1.2 PAST AND CURRENT STATUS OF AI (ORIGIN)

Here is the history of AI during 20th century –

Year	Milestone / Innovation
1923	Karel Čapek play named “Rossum’s Universal Robots” (RUR) opens in London, first use of the word “robot” in English.
1943	Foundations for neural networks laid.
1945	Isaac Asimov, a Columbia University alumnus, coined the term Robotics.
1950	Alan Turing introduced Turing Test for evaluation of intelligence and published Computing Machinery and Intelligence. Claude Shannon published Detailed Analysis of Chess Playing as a search.
1956	John McCarthy coined the term Artificial Intelligence. Demonstration of the first running AI program at Carnegie Mellon University.
1958	John McCarthy invents LISP programming language for AI.
1964	Danny Bobrow’s dissertation at MIT showed that computers can understand natural language well enough to solve algebra word problems correctly.
1965	Joseph Weizenbaum at MIT built ELIZA, an interactive problem that carries on a dialogue in English.
1969	Scientists at Stanford Research Institute Developed Shakey, a robot, equipped with locomotion, perception, and problem solving.
1973	The Assembly Robotics group at Edinburgh University built Freddy, the Famous Scottish Robot, capable of using vision to locate and assemble models.
1979	The first computer-controlled autonomous vehicle, Stanford Cart, was built.
1985	Harold Cohen created and demonstrated the drawing program, Aaron.
1990	Major advances in all areas of AI – (i) Significant demonstrations in machine learning (ii) Case-based reasoning (iii) Multi-agent planning (iv) Scheduling (v) Data mining (vi) Web Crawler (vii) Games (viii) natural language understanding and translation (ix) Vision, Virtual Reality
1997	The Deep Blue Chess Program beats the then world chess champion, Garry Kasparov.
2000	Interactive robot pets become commercially available. MIT displays Kismet, a robot with a face that expresses emotions. The robot Nomad explores remote regions of Antarctica and locates meteorites.

1.3 ARTIFICIAL INTELLIGENCE TECHNIQUES

An AI Technique is a method that exploits knowledge that is represented so that:

- 1) The knowledge captures generalization that share properties, are grouped together, rather than being allowed separated representation.

- 2) It can be understood by people who must provide it. Even though for many programs bulk of the data comes automatically from readings.

Some AI Techniques are: (a) neural network. (b) Fuzzy logic (c) Genetic algorithm
NEURAL NETWORK

Neural network belongs to the family of machine learning algorithm and are inspired by the functioning of neuron in the human brain. They are based on the fact that, given some parameter there is a way to combine them to produce a specific result. The data goes through different layers in which a series of learning rules are applied until reaching the last layer. Where results are compared with the "correct" one and the parameters are adjusted based on the "weight" function given in each rule. Once the network has learned, it can fix its "weights" and run in memory or execution mode.

Artificial Neural Networks (ANNs)

The inventor of the first neurocomputer, Dr. Robert Hecht-Nielsen, defines a **neural network** as "...a computing system made up of a number of simple, highly interconnected processing elements, which process information by their dynamic state response to external inputs."

Basic Structure of ANNs

The idea of ANNs is based on the belief that working of human brain by making the right connections, can be imitated using silicon and wires as living **neurons** and **dendrites**.

The human brain is composed of 86 billion nerve cells called **neurons**. They are connected to other thousand cells by **Axons**. Stimuli from external environment or inputs from sensory organs are accepted by dendrites. These inputs create electric impulses, which quickly travel through the neural network. A neuron can then send the message to other neuron to handle the issue or does not send it forward.

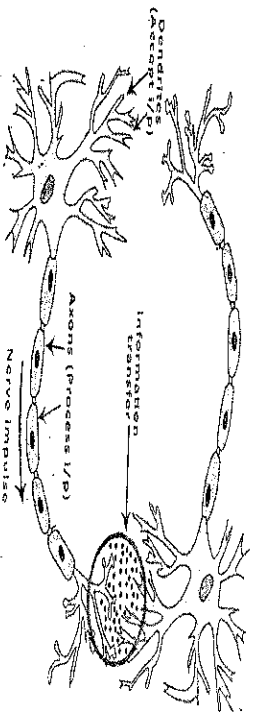


Figure 1: Structure of Human brain

ANNs are composed of multiple **nodes**, which imitate biological **neurons** of human brain. The neurons are connected by links and they interact with each other. The nodes can take input data and perform simple operations on the data. The result of these operations is passed to other neurons. The output at each node is called its **activation** or **node value**.

Each link is associated with **weight**. ANNs are capable of learning, which takes place by altering weight values. The following illustration shows a simple ANN –

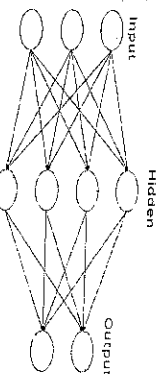


Figure 2: Structure of Artificial Neural Network

Types of Artificial Neural Networks

There are two Artificial Neural Network topologies – **Feed Forward** and **Feedback**.

Feed Forward ANN: -In this ANN, the information flow is unidirectional. A unit sends information to other unit from which it does not receive any information. There are no feedback loops. They are used in pattern generation/recognition/classification. They have fixed inputs and outputs.

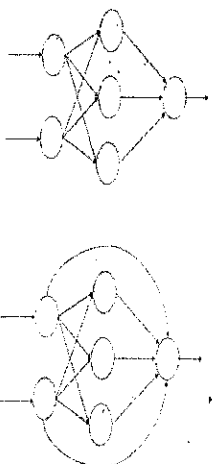


Figure 3: Structure of Feed Forward ANN

Feedback ANN: - Here, feedback loops are allowed. They are used in content addressable memories.

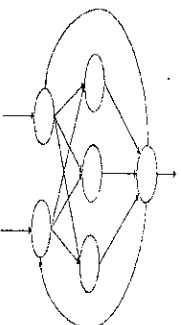


Figure 4: Structure of feedback ANN

Working of ANNs

In the topology diagrams shown above, each arrow represents a connection between two neurons and indicates the pathway for the flow of information. Each connection has a weight, an integer number that controls the signal between the two neurons.

If the network generates a “good or desired” output, there is no need to adjust the weights. However, if the network generates a “poor or undesired” output or an error, then the system alters the weights in order to improve subsequent results.

Application of Neural Networks

They can perform tasks that are easy for a human but difficult for a machine –

- **Aerospace** – Autopilot aircrafts, aircraft fault detection.
- **Automotive** – Automobile guidance systems.
- **Military** – Weapon orientation and steering, target tracking, object discrimination, facial recognition, signal/image identification.
- **Electronics** – Code sequence prediction, IC chip layout, chip failure analysis, machine vision, voice synthesis.
- **Financial** – Real estate appraisal, loan advisor, mortgage screening, portfolio trading program, corporate financial analysis, currency value prediction, document readers, credit application evaluators.
- **Industrial** – Manufacturing, process control, product design and analysis, quality inspection systems, welding quality analysis, paper quality prediction, chemical product design analysis, dynamic modeling of chemical process systems, machine maintenance analysis, project bidding, planning, and management.
- **Medical** – Cancer cell analysis, EEG and ECG analysis, prosthetic design, transplant time optimizer.
- **Speech** – Speech recognition, speech classification, text to speech conversion.

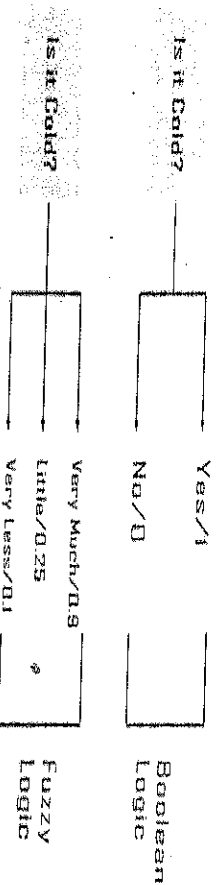
- **Telecommunications** – Image and data compression, automated information services, real-time spoken language translation.
- **Transportation** – Truck Brake system diagnosis, vehicle scheduling, routing systems.
- **Software** – Pattern Recognition in facial recognition, optical character recognition, etc.
- **Time Series Prediction** – ANNs are used to make predictions on stocks and natural calamities.
- **Signal Processing** – Neural networks can be trained to process an audio signal and filter it appropriately in the hearing aids.
- **Control** – ANNs are often used to make steering decisions of physical vehicles.
- **Anomaly Detection** – As ANNs are expert at recognizing patterns, they can also be trained to generate an output when something unusual occurs that misfits the pattern.

FUZZY LOGIC

In artificial intelligence (AI) systems, fuzzy logic is used to imitate human reasoning and cognition. Rather than strictly binary cases of truth, fuzzy logic includes 0 and 1 as extreme cases of truth but with various intermediate degrees of truth.

The idea of fuzzy logic was first advanced by Lotfi Zadeh of the University of California at Berkeley in the 1960s. Zadeh was working on the problem of computer understanding of natural language. Natural language -- like most other activities in life and indeed the universe -- is not easily translated into the absolute terms of 0 and 1. Whether everything is ultimately describable in binary terms is a philosophical question worth pursuing, but in practice, much data we might want to feed a computer is in some state in between and so, frequently, are the results of computing. It may help to see fuzzy logic as the way reasoning really works and binary, or Boolean, logic is simply a special case of it.

✶ **Fuzzy Logic (FL)** is a method of reasoning that resembles **human reasoning**. This approach is similar to how humans perform decision making. And it involves all intermediate possibilities between **YES** and **NO**.



The **conventional logic block** that a computer understands takes precise input and produces a definite output as TRUE or FALSE, which is equivalent to a human being's YES or NO. The Fuzzy logic was invented by Lotfi Zadeh who observed that unlike computers, humans have a different range of possibilities between YES and NO, such as: **CERTAINLY YES, POSSIBLY YES, CANNOT SAY, POSSIBLY NO, CERTAINLY NO**.

Uses of Fuzzy Logic

Generally, we use the fuzzy logic system for both commercial and practical purposes such as:

- It **controls machines and consumer products**
- If not accurate reasoning, it at least provides acceptable reasoning
- It helps in dealing with the **uncertainty in engineering**

Fuzzy Logic Architecture

The fuzzy logic architecture consists of four (4) main parts:

- (i) **Rules** – It contains all the rules and the if-then conditions offered by the experts to control the decision-making system.
- (ii) **Fuzzification** – This step converts inputs or the crisp numbers into fuzzy sets. You can measure the crisp inputs by sensors and pass them into the **control system** for further processing. It splits the input signal into five steps such as-

LARGE POSITIVE, MEDIUM POSITIVE, SMALL, MEDIUM NEGATIVE, LARGE NEGATIVE

- (iii) **Inference Engine** – It determines the degree of match between fuzzy input and the rules. According to the input field, it will decide the rules that are to be fired. Combining the fired rules, form the control actions.

- (iv) **Defuzzification** – The Defuzzification process converts the fuzzy sets into a crisp value. There are different types of techniques available, and you need to select the best-suited one with an expert system.

Applications of Fuzzy Logic

- It is used in the **aerospace field for altitude control** of spacecraft and satellite.
- It is used for **decision making support systems** & personal evaluation in the large business.
- It also controls the pH, drying, chemical distillation process in the **chemical industry**.
- Fuzzy logic is used in **Natural language processing** and various intensive AI.
- It is extensively used in **modern control systems** such as expert systems.
- Fuzzy Logic mimics how a person would make decisions, only much faster. Thus, you can use it with Neural Networks.

Advantages of Fuzzy Logic

- The structure of Fuzzy Logic Systems is **easy and understandable**
- Fuzzy logic is widely used for **commercial and practical purposes**
- It helps to **control machines** and consumer products
- It helps to deal with the **uncertainty in engineering**
- It can **easily modify** it to improve or alter system performance
- **Inexpensive sensors** can be used which helps you to keep the overall system cost & complexity low

Disadvantages of Fuzzy Logic

- It is **not always accurate**. So the results are perceived based on assumptions and may not be widely accepted
- It **cannot recognize machine learning as well as neural network** type patterns
- **Validation & Verification** of a fuzzy knowledge-based system needs **extensive testing** with hardware
- At times, the fuzzy logic is **confused with probability theory**

GENETIC ALGORITHM

The genetic algorithm is a method for solving both constrained and unconstrained optimization problems that is based on natural selection, the process that drives biological evolution. The genetic algorithm repeatedly modifies a population of individual solutions.

The evolution theory is the basis of genetic algorithms. These algorithms use the direct random search method. The developers calculate the fit function to cross the two fittest options to create a favorable child. The

developers gather the population data and then evaluate each member to calculate everyone's fitness. It is estimated by how well each member fits with the desired requirement. Then the developers use various selection methods to keep the best members. Some of the ways are the following:

1. Rank Selection
2. Tournament Selection
3. Steady Selection
4. Roulette Wheel Selection (Fitness Proportionate Selection)
5. Elitism

14 AREA OF APPLICATION OF ARTIFICIAL INTELLIGENCE

AI application in E-commerce: Artificial Intelligence technology is used to create recommendation engines through which you can engage better with your customers. **Fraud Prevention:** Credit card frauds and fake reviews are two of the most significant issues that E-Commerce companies deal with. AI can help reduce the possibility of credit card fraud taking place.

Application of AI in Education: - (i) **Creating Smart Content:** Digitization of content like video lectures, conferences, and textbook guides can be made using Artificial Intelligence. (ii) **Voice Assistants:** Without even the direct involvement of the lecturer, a student can access extra learning material through Voice Assistants. Through this, it reduce printing costs of temporary handbooks.

Application of AI in Lifestyle: (i) **Autonomous Vehicles:** Automobile manufacturing companies like Toyota, Audi, Volvo, and Tesla use machine learning to train computers to think and evolve like humans when it comes to driving in any environment and object detection to avoid accidents. (ii) **Spam Filters:** The email that we use in our day-to-day lives has AI that filters out spam emails sending them to spam or trash folders. The popular email provider, Gmail, has managed to reach a filtration capacity of approximately 99.9%. (iii) **Facial Recognition:** Our favorite devices like our phones, laptops, and PCs use facial recognition techniques by using face filters to detect and identify in order to provide secure access.

Application of AI in Robotic: Robotics is another field where Artificial Intelligence applications are commonly used. Robots powered by AI use real-time updates to sense obstacles in its path and pre-plan its journey instantly. It can be used for: (I) carrying goods in hospitals, factories, and warehouses (II) Cleaning offices and large equipment (III) Inventory management

Application of AI in health care: AI finds diverse applications in the healthcare sector. They are used in healthcare to build sophisticated machines that can detect diseases and identify cancer cells. AI can help analyze chronic conditions with lab and other medical data to ensure early diagnosis. AI uses the combination of historical data and medical intelligence for the discovery of new drugs.

Application of AI in Agriculture: Artificial Intelligence is used to identify defects and nutrient deficiencies in the soil. This is done using computer vision, robotics, and machine learning applications. AI can analyze where weeds are growing. AI bots can help to harvest crops at a higher volume and faster pace than human laborers.

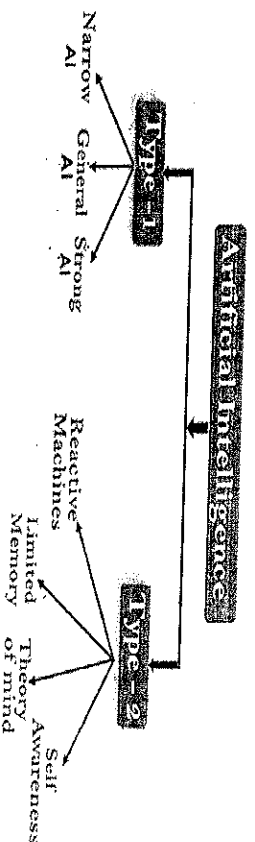
Application of AI in Gaming: AI can be used to create smart, human-like NPCs to interact with the players. It can also be used to predict human behavior using which game design and testing can be improved. The Alien Isolation game released in 2014 uses AI to stalk the player throughout the game. The game uses two Artificial Intelligence systems - 'Director AI' that frequently knows your location and the 'Alien AI,' driven by sensors and behaviors that continuously hunt the player

Application of AI in Automobiles: Automobile manufacturing companies like Toyota, Audi, Volvo, and Tesla use machine learning to train computers to think and evolve like humans when it comes to driving in any environment and object detection to avoid accidents.

Application of AI in Social media: - **Instagram:** On Instagram, AI considers your likes and the accounts you follow to determine what posts you are shown on you're explore tab. **Facebook:** AI is also used along with a tool called Deep Text. With this tool, Facebook can understand conversations better. It can be used to translate posts from different languages automatically. **Twitter:** AI is used by Twitter for fraud detection, for removing propaganda, and hateful content. Twitter also uses AI to recommend tweets that users might enjoy, based on what type of tweets they engage with.

1.5 CLASSIFICATION OF ARTIFICIAL INTELLIGENCE

Artificial Intelligence can be divided in various types, there are mainly two types of main categorization which are based on capabilities and based on functionality of AI.



AI TYPE-1: BASED ON CAPABILITIES

1. **Weak AI or Narrow AI:** - Narrow AI is a type of AI which is able to perform a dedicated task with intelligence. Narrow AI cannot perform beyond its limitations, as it is only trained for one specific task. Eg. Apple Series, IBM's Watson supercomputer, playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition.
2. **General AI:**-General AI is a type of intelligence which could perform any intellectual task with efficiency like a human. The idea behind the general AI to make such a system which could be smarter and think like a human by its own.
3. **Super AI:** - Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties. Some key characteristics of strong AI include capability include the ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate by its own.

AI TYPE-2: BASED ON FUNCTIONALITY

Below are the various types of AI:

1. **Purely Reactive Machines:** These machines do not have any memory or data to work with, specializing in just one field of work. For example, in a chess game, the machine observes the moves and makes the best possible decision to win.
2. **Limited Memory:** These machines collect previous data and continue adding it to their memory. They have enough memory or experience to make proper decisions, but memory is minimal. For example, this machine can suggest a restaurant based on the location data that has been gathered.
3. **Theory of Mind:** This kind of AI can understand thoughts and emotions, as well as interact socially. However, a machine based on this type is yet to be built.
4. **Self-Aware:** Self-aware machines are the future generation of these new technologies. They will be intelligent, sentient, and conscious.

MODULE TWO

PROBLEM SOLVING TECHNIQUE

2.1 PROBLEM SOLVING TECHNIQUES

In computer science, the term "problem-solving" refers to artificial intelligence methods, which may include formulating, ensuring appropriate, using algorithms, and conducting root-cause analyses that identify reasonable solutions.

Problem Solving in games such as "Sudoku" can be an example. It can be done by building an artificially intelligent system to solve that particular problem. To do this, one needs to define the problem, understand the constraints, and then generating the solution by keeping the conditions in mind.

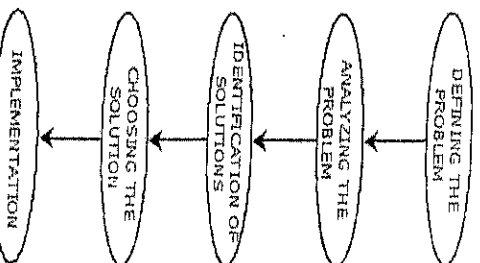
Some of the most popularly used problem solving with the help of artificial intelligence are:

- (1) Chess.
- (2) Travelling Salesman Problem.
- (3) Tower of Hanoi Problem.
- (4) Water-Jug Problem.
- (5) N-Queen Problem.

Problem: Problems are the issues which comes across any system. A solution is needed to solve that particular problem.

Steps To Solve Problem Using Artificial Intelligence

The process of solving a problem consists of five steps. These are:



Problem Solving in Artificial Intelligence

1. **Defining The Problem:** The definition of the problem must be included precisely. It should comprise the possible initial as well as final situations which should result in acceptable solution.
2. **Analyzing The Problem:** Analyzing the problem and its requirement must be done as two factors, as they have immense impact on the resulting solution.
3. **Identification of Solutions:** This phase generates reasonable amount of solutions to the given problem in a particular range.

4. Choosing a Solution: From all the identified solutions, the best solution is chosen based on the results produced by respective solutions.
5. Implementation: After choosing the best solution, its implementation is done.

2.2 APPROACHES TO PROBLEM FORMULATION.

Problem formation is the step in problem definition that is used to understand and decide a course of action that needs to be considered to achieve a goal. If there is more than one way an agent can reach its goal, then it causes complexity in terms of actually achieving the goal as there would be too many steps and paths that the AI entity can take to reach the goal that it causes confusion and a massive drop in the efficiency at the same time.

Problem formulation can be done in many steps such as the definition of the initial state of the agent, determining possible actions that the agent can take, creation of transition models to describe the actions of the agent. Testing the goal and determining the path cost function are also important complements in determining the problem formulation approach.

A problem-solving approach is a technique people use to better understand the problems they face and to develop optimal solutions. The approaches empower people to devise more innovative solutions by helping them overcome old or binary ways of thinking

There are basically three types of problem in artificial intelligence:

1. **Ignorable:** In which solution steps can be ignored.
2. **Recoverable:** In which solution steps can be undone.
3. **Irrecoverable:** Solution steps cannot be undo.

There are 3 main approaches to solving a problem:

- Intuitive:- having the ability to know or understand things without any proof or evidence
- Analytical:- examining or to examine things in detail, in order to discover more about them.
- Experimental:- a study conducted with a scientific approach using two sets of variables.

2.3 ARTIFICIAL INTELLIGENT PROBLEM

Problems that AI helps to solve in companies:

- Customer support
- Data analysis
- Demand forecasting
- Fraud
- Image and video recognition
- Predicting customer behavior
- Productivity

The Artificial Intelligence Problems include :-

1. Job Loss Problem

Job loss concerns related to **Artificial Intelligence** have been the subject of numerous business cases and academic studies. As per an Oxford Study, more than 47% of American jobs will be under threat due to automation by the mid-2030s. According to the World Economic Forum, Artificial Intelligence automation will replace over 75 million jobs by 2022

2. Safety Problem

Safety issues associated with Artificial Intelligence have always stirred up a lot of furor. When experts like Elon Musk, Stephen Hawking, and Bill Gates, among others, express concern about AI safety, we should pay heed to its safety issues. Artificial Intelligence has encountered instances where Twitter Chatbot began spewing abusive and Pro-Nazi sentiments, and Facebook AI bots started interacting in an incomprehensible language, resulting in the project's shutdown.

3. Trust-Related Problem

As Artificial Intelligence algorithms become more powerful by the day, it also brings several trust-related issues on their ability to make fair decisions for the betterment of humankind. With AI slowly reaching human-level cognitive abilities, the trust issue becomes all the more significant.

4. Computation Problem

Artificial Intelligence algorithm analyzes the humongous data that require immense computational power. So far, the problem has been dealt with with the help of Cloud Computing and Parallel Processing.

MODULE THREE

HOW TO RELATE PROBLEM-SOLVING TO ARTIFICIAL INTELLIGENCE

The concept of relating problem-solving to Artificial intelligence.

MODULE FOUR

MACHINE LEARNING

4.0 BACKGROUND OF MACHINE LEARNING

(Herbert Simon, 1983) Learning denotes changes in a system that enable a system to do the same task more efficiently the next time. (Marvin Minsky, 1986) Learning is making useful changes in the workings of our minds. (Ryszard Michalski, 1986) Learning is constructing or modifying representations of what is being experienced.

Learning is the process of acquiring new understanding, knowledge, behavior, skilled, value, attitude and Preference. The ability to learn is processed by humans, animals and some machine. There is also evidence for some kind of learning in certain plants.

Learning in AI is the process that improves the knowledge environments. Machine learning (ML) is a branch of AI data, identify patterns and make decisions with minimal human intervention.

We require Machine Learning to:-

- (i) Understand and improve efficiency of human learning.
- (ii) Discover new things or structure that is unknown to humans.
- (iii) Fill in skeletal or incomplete specifications about a domain.

4.1 FORMS OF LEARNING

There are three general categories of learning that AI machine learning utilities to actually learn. They are:

- (a) Based on their learning pattern (b) Based on similarity of function

Based On Their Learning Pattern

- (i) **Supervised Learning:** - The Machine has a “teacher” who guides it by providing sample inputs along with the desired output.
- (ii) **Unsupervised Learning:-** This is the most important and Most difficult type of learning based on prediction. Eg Babies learn early in life.
- (iii) **Reinforcement Learning:** is a computer program that interact with a dynamic environment in which it must perform a certain goal. Eg. Driving a vehicle.

Based on Similarities of Function

- (a) **Regression:** is the process that is concerned with identifying the relationship between the target output variable and the input features to make prediction about the new data. Top six regression algorithm are:
 - (i) Simple learning regression.
 - (ii) Lasso Regression.
 - (iii) Logistic Regression.
 - (iv) Multivariate regression.
 - (v) Multiple regression.
- (b) **Decision Tree:** these are tree-based model constructed on the decision made by examining the value of the attributes.
 - (i) Classification and regression tree.
 - (ii) Conditional decision Tree.
 - (iii) C4.5 & C5.0
 - (iv) Decision stump.
 - (v) Chi-Squared automatic Interaction. Detection.

(c) Artificial Neural Network (ANN):- These relies and find it based from the biological neuron in the human brain (classification and regression problem).

(i) Basic function Neural Network.(ii) Perception. (iii) Multilayer perception. (iv) Stochastic Gradient Decent. (v) Back propagation. (vi) Hopfield Neural Network.

(d) Deep learning:- they are Modernized Version of ANN that can handle very large and complex data base of labeled data. (Handle text, images, audio and video data).

(i) Convolutional Neural Network (ii) Recurrent Neural Network (iii) Deep Boltzman Machine (iv) Auto encoders Deep belief Neural Network (v) Long Short-term memory Neural Network

(e) Dimensionality Reduction:- it exploit the intrinsic structure of data. In an unsupervised manner to express data using reduced information set.

(i) Principal component analysis (ii) Principal component Regression (iii) Linear Decrement analysis

(iv) Quadratic Discrement Analysis (v) Flexible Discrement analysis (vi) Sammon mapping

(f) Instance-Based:- these belong to the family of learning that measures new instances of the problem with those in the training data to find out a best match and make a prediction.

(i) K-Nearest neighbor (ii) Learning vector organization (iii) Self Organizing Map (iv) Locally weighted learning (v) Support vector machine

(g) Regularization:- it refers to techniques of regularizing the leaking process from a particular set of features.

(i) Ridge regression (ii) Least angle regression (LARs) (iii) Least absolute shrinkage and selection operator (Lasso)

(h) Bayesian Algorithms:- these algorithm applies the bayes theorem for classification and regression. Problem

(i) Naive Bayes (ii) Gaussain Naive Bayes (iii) Multinomial Noise Bayes (iv) Bayesian belief N/N (v) Bayesian N/N (vi) Average one – depended Estimator

4.2 CHARACTERISTICS OF LEARNING IN AI.

The major characteristics of learning are:-

- i) Rate Learning:- the learning is by memorizing. I.e. by saving knowledge so that it can be used again.
- ii) Induction:- it involves the process of learning by an example where a system tries to induced a general rule from a set of observed instances.
- iii) Analogy:- it can recognize similarities in information already stored. It can determine the correspondence between two different representations.
- iv) Reinforcement:- it assigns rewards positive or negative at the end of a sequence of steps and it learn which creation are good or bad.
- v) Genetic algorithm:- it's a part of a evolutionary compiling it's a way of solving problem by mimicking a problem, nature, uses selection, crossover, mutation and acceptance to evolve a solution to a problem.

MODULE FIVE

PROBLEM SOLVING BY SEARCHING STRATEGIES

Problem-solving is commonly known as the method to reach the desired goal or find a solution to a given situation. In computer science, problem-solving refers to artificial intelligence techniques, including various techniques such as forming efficient algorithms, heuristics, and performing root cause analysis to find desirable solutions. Problem-solving in Artificial Intelligence usually refers to researching a solution to a problem by performing logical algorithms, utilizing polynomial and differential equations, and executing them using modeling paradigms. There can be various solutions to a single problem, which are achieved by different heuristics. Also, some problems have unique solutions. It all rests on the nature of the given problem.

5.2 Stages of Problem Solving In AI

- **Goal** – In this phase as soon as a crisis appears, the AI agent puts a goal or a mark. This needs the agent to thoroughly examine and clarify the issue.
- **Problem Description** – This is one major stage of problem-solving. Whenever a problem occurs, then the agent chooses what measures must be carried out to run to the developed goal.
- **Describing the State** – A state area can be described as a group of all the accurate conditions in which an agent can be joined when discovering a key to the crisis.
- **Specifying Primary State** – For an agent to begin cracking the issue, it must begin from a state. The primary state from where the agent begins performing is directed to the primary state.
- **Collect Details** – Now the agent collects data and utilizes the data needed by it to fix the issue. These details will be collected with one-time incidents as well as present pieces of knowledge.
- **Designing the Changes** – Some issues are undersized. But most of the time issues will be such where sound planning and implementation are required. Hence this needs appropriate data structures and managing processes well in advance.

5.3 Method of Executing Algorithm In problem solving

There are four methods in which the execution of an algorithm is calculated. These are as follows:

- **Totally** – Totally calculates the algorithm's assurance to discover the solution for the issue if there is any solution for it.
- **Optimality** – This step is utilized to calculate the tracking methods which discover an optimal solution to the presented issue.
- **Duration Complexity** – This calculates the portion of time the algorithm abides to run till the key for the assigned issue.
- **Space Intricacy** – This is the standard that is employed to specify the quantity of area (in memory) that the algorithm needs to execute the quest.

5.4 Searching Algorithms

Searching is one of the primary methods of solving any problem in AI. Rational agents or problem-solving agents use these searching algorithms to find optimal solutions. These problem-solving agents are often goal-based and utilize atomic representation. Moreover, these searching algorithms possess completeness, optimality, time complexity, and space complexity properties based on the quality of the solution provided by them.

5.5 Types of Searching Algorithms

1. Informed Search
2. Uninformed Search

Informed Search:- These algorithms use basic domain knowledge and comprehend available information regarding a specified problem as a guideline for optimal solutions. The solutions provided by informed search algorithms are more efficient than uninformed search algorithms.

Types of informed Search Algorithms

There are following main two types of informed search algorithms:

- (i) Greedy Search
- (ii) A * Search

Uninformed Search:- These algorithms do not have the privilege of using basic domain knowledge, such as the desired goal's closeness. It contains information regarding traversing a tree and identifying leaf and goal nodes. Uninformed search also goes by the name of blind search because while traversing, there is no specific information about the initial state and test for the goal. This search goes through every node till reaching the desired destination.

Types of Uninformed Searching Algorithms

1. Breadth-First Search
2. Depth First Search:-
3. Uniform Cost Search
4. Iterative Deepening Depth First Search
5. Bidirectional Search

CHAPTER SIX

ARTIFICIAL INTELLIGENT AGENTS

6.1. ARTIFICIAL INTELLIGENT AGENT

Learning Agents (An agent) is an entity that is capable of **perceiving** and **do action**. An agent can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**.

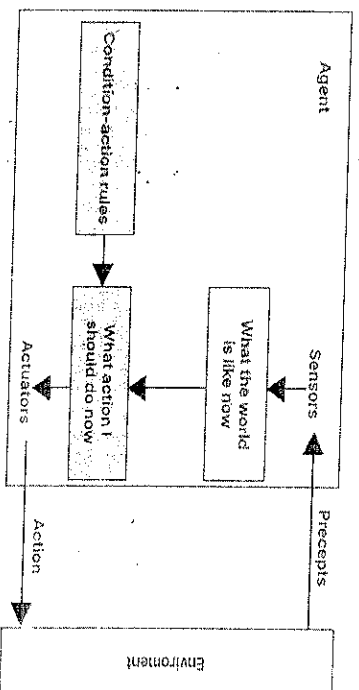
In artificial intelligent an agent is a computer program or system that is design to perceive it environmental, makes decision and takes action to achieve a specific goal or set of goals.

The agent operate autonomously, meaning it's not directly controlled by a human operation.

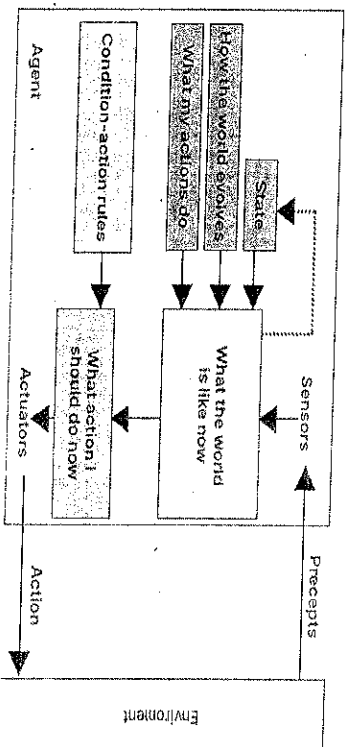
6.2. TYPE OF AGENT

Based on the degree of perceived intelligent and capacity. Agent can be divide into five type which are:

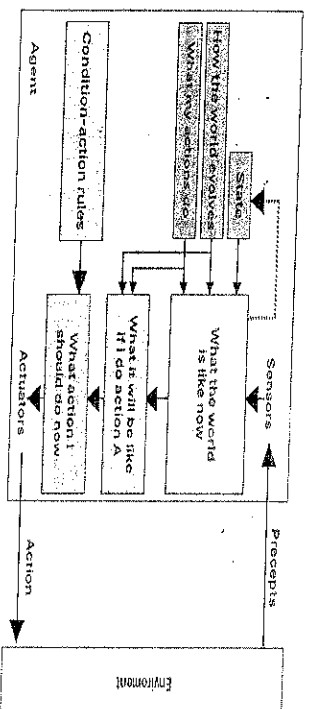
- (i) Simple reflex agent (ii) Model based agent (iii) Goal based agent (iv) Utility agent (v) Learning agent
- i. **Simple reflex agent**: is a simple type of agent which work on the bass of current percept and not based on the rest of the percept history.



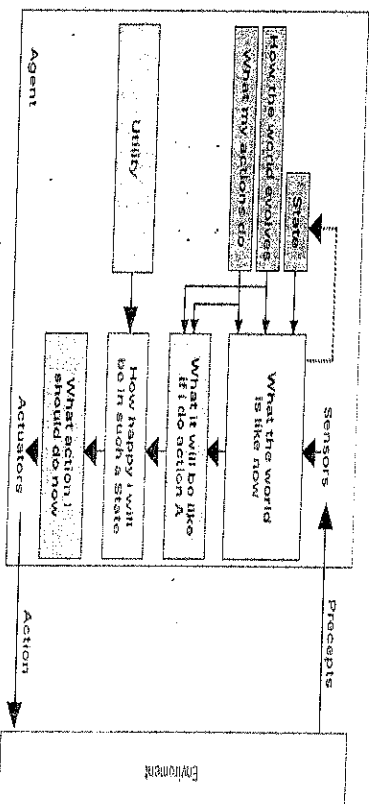
- ii. **Model based agent**: Model based agent utilizes condition action rule where it work by finding a rule that will allow the condition, which is based on the current situation to be satisfied.



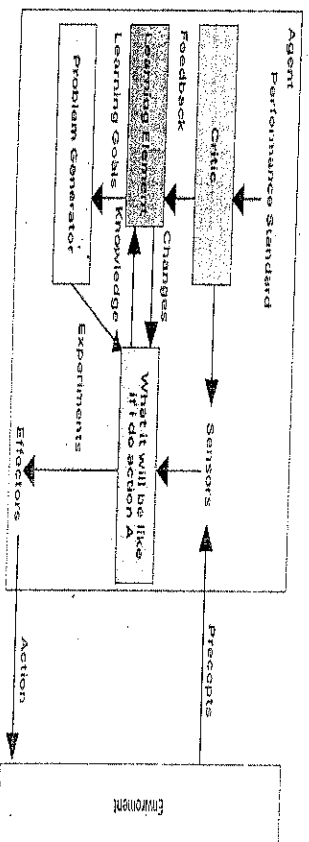
- iii. **Goal based agent**: this type of agent takes decision on the basis of its goal or desirable situation. So that it can choose such an action that can achieve the goal required.



- iv. **Utility agent:** utility agent have their end user as their building blocks and is used when best action and decision needs to be taken from multiple alternatives.



- v. **Learning agent:** learning agent as the name suggest, has the capability to learn from past experiences and takes action or decision based on learning capabilities.



6.3 STRUCTURE OF INTELLIGENT AGENT

The structure of an intelligent agent is a combination of agent function, architecture and agent program. It can be viewed as.

AGENT=ARCHITECTURE + AGENT PROGRAM

- i. **Architecture:** is the machine on which the agent executes its action. It is essentially a device with embedded actuators and sensors. Example: Autonomous car which has various motion and GPS sensor attached to them and actuators based on the inputs and in actual driving.
- ii. **Agent Functions:** it help in mapping all the information it has gathered from the environment in to the action.
- iii. **Agent Program:** it performs the execution of the agent function. The execution happens on top of the agent architecture and produce the desired function.

6.4 PARAMETERS FOR IDEAL AGENT.

Intelligent Agent (Learning Agent) Agent is an entity that is capable of perceiving and do action. In computer science an **agent** is a software agent. In artificial intelligence/Engineering, the term used for agent is an intelligent agent. Learning is an important feature of "Intelligence".

Percept : agent's perceptual inputs

Percept sequence: history of everything the agent has perceived

Agent function : describes agent's behavior

Agent program : Implements agent's function

Machine uses (processor)

6.5 COMPONENTS OF A LEARNING SYSTEM (LEARNING AGENT)

Learning Agent consist of four main components:

- (I) **Performance Element:** The Performance Element is the agent itself that acts in the world. It takes in percepts and decides on external actions.
- (II) **Learning Element:** It responsible for making improvements, takes knowledge about performance element and some feedback, determines how to modify performance element.
- (III) **Critic:** Tells the Learning Element how agent is doing (success or failure) by comparing with a fixed standard of performance.
- (IV) **Problem Generator:** Suggests problems or actions that will generate new examples or experiences that will aid in training the system further.

Example: Automated Taxi on city roads

Performance Element: Consists of knowledge and procedures for driving actions. e.g., turning, accelerating, braking are performance element on roads.

Learning Element: Formulates goals. e. g., learn rules for braking, accelerating, learn geography of the city.

Critic: Observes world and passes information to learning element. e. g. quick right turn across three lanes of traffic, observe reaction of other drivers.

Problem Generator: Try south city road.

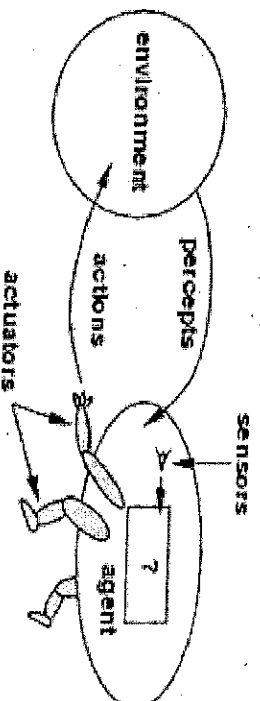
6.6. AGENT ENVIRONMENT.

An agent is anything that can perceive its environment through sensors and acts upon that environment through effectors.

A Human Agent: A human agent has sensory organs such as eyes, ear, nose, tongue and skin parallel to the sensors and order organs such as hands, legs, mouth for effector.

A Robotic Agent: A robotic agent replace cameras and infrared range finders for the sensors and various motors and actuators for the effector.

A Software Agent: A software Agent has encoded bit strings as it program and actions. In computer science an **agent is a software agent** that assists users and acts in performing computer-related tasks.



Environment	Sensors	Actuators
Agent		
Human agent	Eyes, ears, etc	Leg. hands, mouth
Robotic agent	Cameras, IR range finders	Motors
Software agent	Key stroke, File contents	Displays to screen, Write files

6.7. AGENT TERMINOLOGY

- ❖ **Performance measure of agent:** it is the criteria which determine how successful an agent is.
- ❖ **Behavior of agent:** it is the action that agent performs after any giving sequence of percepts.
- ❖ **Percept:** it is agents perceptual input at a given instance.
- ❖ **Agent Function:** it is a map from the percept sequence to an action.
- ❖ **Rationality:** is nothing but status of being reasonable, sensible and having good sense of judgment.
- ❖ **Ideal rational agent:** is the one which is capable of doing expected action to maximize its performance measure, on the basis of: (i) it's percent sequence (ii) its built-in knowledge base

6.8 CHARACTERISTIC OF AGENT ENVIRONMENT

- i. **Discrete/Continuous:** if there are limited number of distinct, clearly defined, state of the environment, the environment is discrete. E.g. chess otherwise it is continuous e.g. driving.
- ii. **Observable/Partially Observable:** of it's possible to determine the complete state of the environment at each, time point from the percept. It's observable: otherwise it's only partially observable.
- iii. **Static/Dynamics:** if the environment does not change while an agent is acting, then its static otherwise it is dynamic.
- iv. **Single Agent/Multiple Agent:** the environment may contain other agents which may be of the same or different kind as that of the agent.
- v. **Accessible/In Accessible:** if the agent sensory apparatus can have access to the complete state of the environment, then the environment is accessible to that agent.
- vii. **Deterministic/Non-deterministic:** if the next state of the environment is completely determines by the current state and the actions of the agent, then the environment is deterministic otherwise it is non deterministic.
- viii. **Episodic/Non-episodic:** in an episodic environment each episode consist of the agent perceiving and then acting the quality of it action depends just on the episode itself. Subsequent episode do not depends on the action in the previous episode.

CHAPTER SEVEN

DEEP NEURAL NETWORK

7.1 DEEP LEARNING

Deep Learning is a method in AI that teaches computer to process data in a way that is inspired by the human brain. Deep learning model can recognize complex pattern in pictures, text, sounds and other data to produce accurate insight and prediction.

7.2 HISTORICAL BACKGROUND OF DEEP LEARNING

The history of deep learning can be traced back to 1943, when Walter Pitts and Warren McCulloch creates a computer model based on the neural network of the human brain. They used a combination of algorithms and mathematics they called “threshold logic” to mimic the thought process.

Henry J. Kelley is given credit for developing the basics of a continuous back propagation model in 1960. Stuart Dreyfus develops a simpler version based only on the chain rule.

The earliest efforts in developing deep learning algorithm came from Alexey Grigory Erick Wakhneoko (developed the group method of Data Handling) and Valentin Grigorich Lapa (author of cybernetics and forecasting techniques) in 1965.

CNN was used by Kunihiko Fukushima. In 1979 he design Neural Network with multiple pooling and convolutional layer.

In 1989, Yann Lecun demonstrate back propagation at bell labs. He combine CNN and back propagation to read “hand written” digits.

In 1995, Dana Cortes and Viadmir Vapnik developed the support vector machine. (System for mapping and recognizing similar data).

In 2000 the Vanishing Gradient Problem appeared was discovered.

In 2009 Fei-Fei, an AI preofessor at stand ford launched “ImageNet”

By 2011, the speed of GPUs had increased significantly, making it possible to train CNN “without the layer-by-layer pre training.

In 2012 Google Brain released the result of an unusual project “The cat experiment supervised learning”

The Generative Adversarial NN (GAN) was introduced in 2014, Lan Good fellow.

1. Why is it called deep learning?

Deep learning is a class of machine learning algorithms that use a deep neural network. A deep neural network is an artificial neural network with multiple hidden layers.

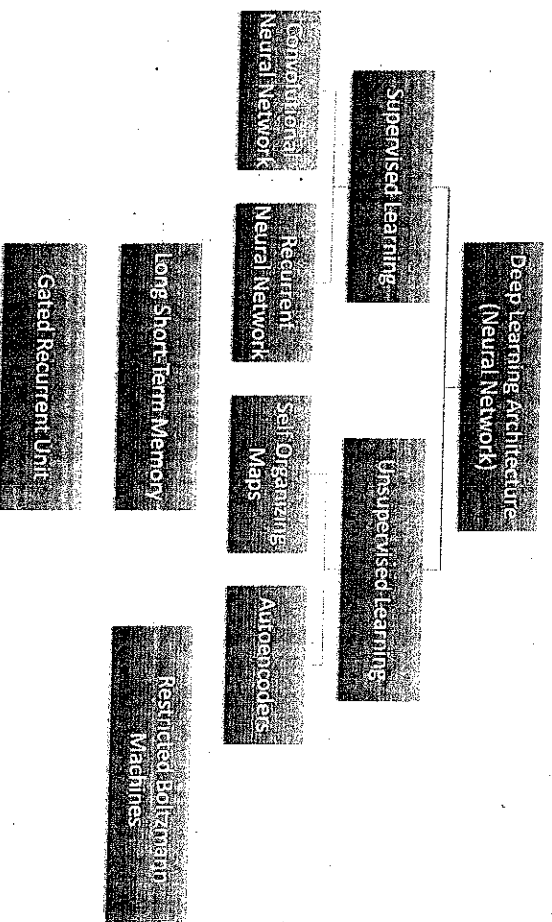
2. What is the difference between AI and deep learning?

Artificial intelligence is a field of computer science focused on creating intelligent machines that reason, learn, and act autonomously. Deep learning is a subset of AI that uses a neural network to learn from data.

3. Why is deep learning so powerful?

Deep learning is powerful because it can learn complex patterns in data. It is possible because deep learning networks can learn multiple levels of representation.

7.3 DEEP LEARNING ARCHITECTURE



Supervised deep learning: - Supervised learning refers to the problem space wherein the target to be predicted is clearly labelled within the data that is used for training.

Convolutional neural networks: - A CNN is a multilayer neural network that was biologically inspired by the animal visual cortex. The architecture is particularly useful in image-processing applications.

Recurrent neural networks: - The RNN is one of the foundational network architectures from which other deep learning architectures are built.

LSTM networks: - The LSTM was created in 1997 by Hoch Reiter and Schmidhuber, but it has grown in popularity in recent years as an RNN architecture for various applications.

GRU networks: - In 2014, a simplification of the LSTM was introduced called the gated recurrent unit. This model has two gates, getting rid of the output gate present in the LSTM model.

7.4 DEEP LEARNING APPLICATIONS

Deep learning is a powerful tool that can be used for a variety of applications. Some popular deep-learning applications include:-

1. Self-Driving Cars
2. News Aggregation and Fraud News Detection
3. Natural Language Processing
4. Virtual Assistants.
5. Entertainment
6. Visual Recognition.
7. Fraud Detection
8. Healthcare
9. Personalization's
10. Detecting Developmental Delay in Children
11. Colorizations of Black and White images
12. Adding sounds to silent movies

13. Automatic Machine Translation
14. Automatic Handwriting Generation
15. Automatic Game Playing
16. Language Translations.
17. Pixel Restoration
18. Photo Descriptions
19. Demographic and Election Predictions
20. Deep Dreaming
21. Chatbots
22. Composing Music.
23. Image Coloring
24. Robotics
25. Image Captioning.
26. Advertising

7.5 DIFFERENCE BETWEEN NEURAL NETWORK AND DEEP LEARNING.

	Neural Network	Deep learning
Definition	A neural network is a model of neurons inspired by the human brain. It is made up of many neurons that are interconnected with each other.	Deep learning NNs are distinguished from NN based on their depth or number of hidden layer.
Architecture	1. Recurrent neural network 2. Symmetrically connected neural 3. Neurons	1. Unsupervised pretrained NNs 2. CNN 3. Mother Board
Structure	Connection and weight propagation function learning rate.	PSU RAM Processor
Time & accuracy	- It generally takes less time to train them - They have lower accuracy than Deep learning systems.	- It generally takes more time to train them - They have high accuracy than Neural Network
Performance	- It gives low performance compared to DL N/W	- It gives high performance compared to NNs
Task interpretation	Your task is poorly interpreted by a NN	The D learning N/W more effectively perceives yours task
Application	Classification, pattern, recognition, predictor, and analysis, clustering decision making, machine learning, deep learning.	Pattern recognition, speech recognition, Neural Language processing, computer games, self-driving car, social N/w filtering etc.
Critique	Train Problems, Hardware problem, real world problem	Theory errors, cyber threats

MODULE EIGHT

UNDERSTAND MODEL PERFORMANCE MEASURES

Introduction

The model performance in a classification problem is assessed through a confusion matrix. The elements of the confusion matrix are utilized to find three important parameters named accuracy, sensitivity, and specificity. The prediction of classes for the data in a classification problem is based on finding the optimum boundary between classes. Based on the values of accuracy, sensitivity, and specificity one can find the optimum boundary. This article explains the relation between sensitivity, specificity, and accuracy and how together they can help to determine the optimum boundary.

Most model-performance measures are based on the comparison of the model's predictions with the (known) values of the dependent variable in a dataset. For an ideal model, the predictions and the dependent-variable values should be equal. In practice, it is never the case, and we want to quantify the disagreement.

- **Confusion Matrix.** A confusion matrix is a table that is used to define the performance of a classification algorithm. A confusion matrix visualizes and summarizes the performance of a classification algorithm.

Confusion Matrix

In any machine learning model, we usually focus on accuracy. But if you are dealing with a classification problem, you also need to worry about the percentage of correct classification and misclassification. So we need a mechanism that not only provides accuracy but also helps in estimating correct classification and misclassification. The confusion matrix serves this purpose. It is an $N \times N$ matrix which helps to evaluate the performance of machine learning model for classification problem. Figure 1 presents the confusion matrix for the binary class classification problem.

		Actual Values	
Predicted Values	Positive	TP	FP
	Negative	FN	TN

Figure 1

TP = True Positive
TN = True Negative
FP = False Positive
FN = False Negative

From the confusion matrix Accuracy, Sensitivity and Specificity is evaluated using the following equations

Accuracy Formula

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

Sensitivity Formula

$$Sensitivity = \frac{TP}{TP + FN}$$

Specificity Formula

$$Specificity = \frac{TN}{TN + FP}$$

Example - Calculate Confusion Matrix

We will take a simple binary class classification problem to calculate the confusion matrix and evaluate accuracy, sensitivity, and specificity. Diabetes in the patient is predicted based on the data of blood sugar level.

Step-By-Step Approach to Calculate the Confusion Matrix

Step 1- Import Data.

Step 2: Feature Engineering

Step 3: Split data into train and test

Step 4: Build a binary classification model

Step 5: Predict Diabetes

Step 6: Confusion Matrix and Accuracy

Step 7: Variation with Cut-Off

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad Sensitivity = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

Replacing TP and TN from Sensitivity and Specificity equation-

$$Accuracy = \frac{Sensitivity[TP + FN] + Specificity[TN + FP]}{TP + FP + FN + TN}$$